THE POSTS AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY

DEPARTMENT OF INFORMATION TECHNOLOGY 1

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FINAL REPORT

Business Intelligence

CHỦ ĐỀ: PHÂN TÍCH DỮ LIỆU MẠNG XÃ HỘI CHO DOANH NGHIỆP

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Thank you to our group members for your support, sharing, and for overcoming challenges together throughout the process. We hope that this report will achieve a positive result, reflecting the hard work our group has put in.

Thank you sincerely!

MỤC LỤC

I. INTRODUCTION TO THE PROBLEM

In the context of the growing social network, users are constantly generating large amounts of data in the form of articles, comments, and reviews. These data contain a wealth of valuable information about users' feelings, perspectives, and behaviors.

Reddit is a popular social networking platform where users engage in discussions on a variety of topics. Analyzing data from Reddit helps a business or organization:

- Better understand user feedback.
- Satisfaction ratings.
- Detect potential trends and issues.
- Detect keywords that are searched for a lot.
- In this problem, the team builds a Business Intelligence system to:
- Analyze user sentiment.
- Evaluate the distribution of ratings, keywords are of interest.
- Observe data trends over time.
- Visualize data with dashboards.

II. DATASET DESCRIPTION

The dataset used in the article is the comments data from the background

Reddit platform. This is a form of user-generated content, which directly reflects the opinions and feelings of users.

The dataset used is comment data from Reddit. The data is initially in .pkl format, then processed by python into a .csv file for import into MySQL and Power BI.

1. Data Characteristics:

+ Large-scale data (approximately 50,000 lines).

+ Dạng text (unstructured data).

+ Analysis fields such as sentiment or rating are not available.

2. Main Attributes:

+ Text: the content of the user's comment.

+ Subreddit: topic or community.

+ Rating: rating score.

+ Sentiment : cảm xúc (positive, neutral, negative).

+ Date: time (created to serve enemy analysis).

III. DATA WAREHOUSE/DATABASE DESIGN

In this article, the team uses MySQL to store data after it has been processed with python. The data is stored in the reddit_bi database and the main table reddit_reviews,reddit_keywords:

Column	Data types	Description
Review_id	INT AUTO_INCREMENT PRIMARY KEY	Mã review
Text_content	TEXT	Comments
Subreddit	VARCHAR(255)	Subreddit Threads
Sentiment	VARCHAR(50)	Emotional Labels
Rating	INT	Review Score
Review_date	DATETIME	Review Period
subreddit	TEXT	Agent Review
sum of count by keywords	VARCHAR(255)	Compare occurrences
word cloud	TEXT	Filter by search

1. ETL Process

The ETL process consists of 3 main steps: extract, transform, and load. 1.Extract

The initial data is read from the redditDataset.pkl file using python.

2. Transform

In this step, the data processed includes:

+ Remove blank data.

+ Eliminate duplicate data.

. + Create more rating columns.

+ Create more sentiment columns.

+ Create more date columns.

+ Export data to reddit_emotion.csv and reddit_keywords files.

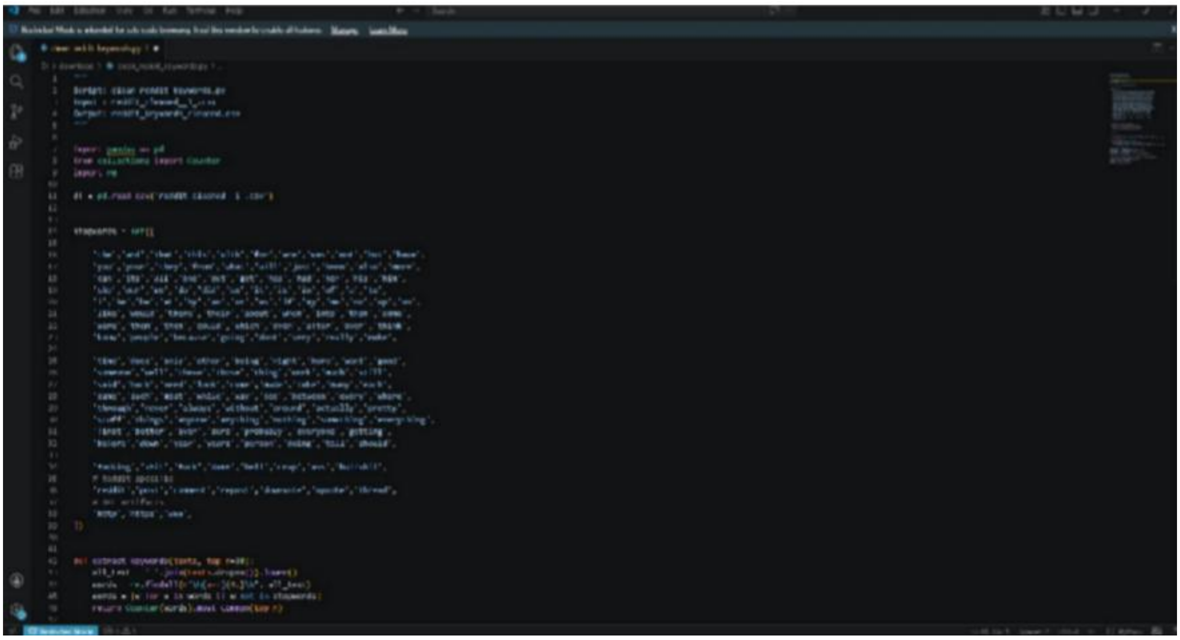
3. Load

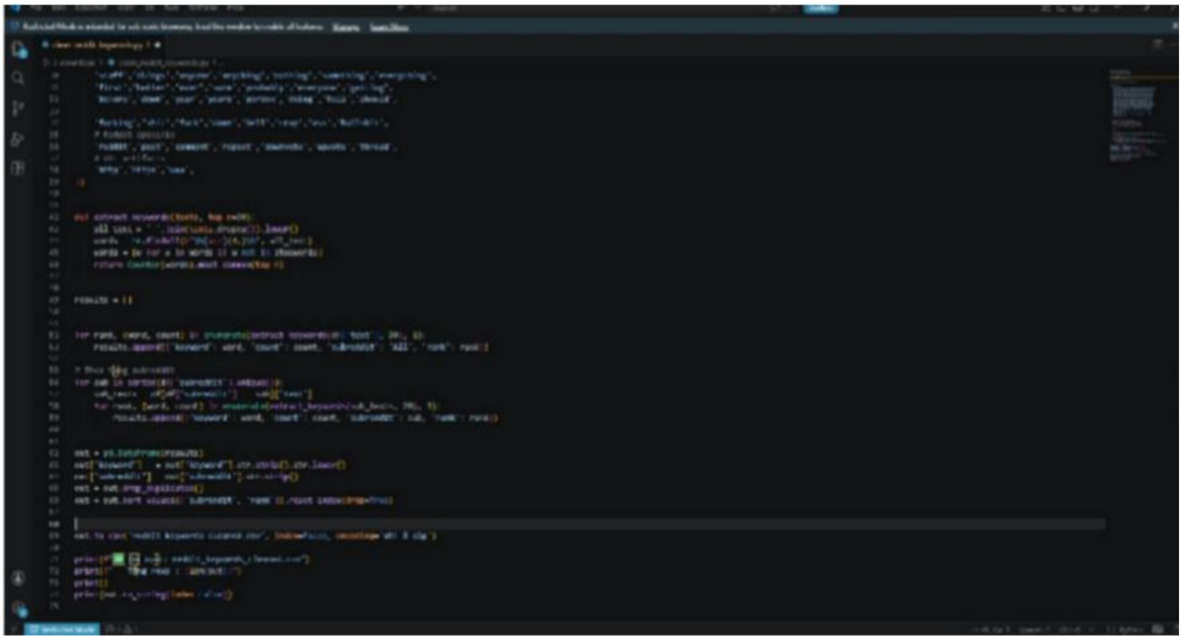
The processed data is imported into MySQL using MySQL Workbench. The data is then fed into power BI to build dashboards.

4. Python code data processing; Reddit Emotion Data Cleaning

```python
import pandas as pd
import numpy as np

# Load dữ liệu Reddit từ file pkl
df = pd.read_pickle("redditDataset.pkl")

# Kiểm tra dữ liệu ban đầu
print(df.head())
print(df.info())

# Làm sạch dữ liệu
df = df.dropna()
df = df.drop_duplicates()

# Tạo rating giả lập từ 1 đến 5
np.random.seed(42)
df["rating"] = np.random.choice(
    [1, 2, 3, 4, 5],
    size=len(df),
    p=[0.25, 0.10, 0.15, 0.15, 0.35]
)

# Tạo sentiment dựa trên rating
def classify_sentiment(rating):
    if rating >= 4:
        return "Positive"
    elif rating == 3:
        return "Neutral"
    else:
        return "Negative"

df["sentiment"] = df["rating"].apply(classify_sentiment)

# Tạo dữ liệu thời gian trong khoảng 2024-2025
date_range = pd.date_range("2024-01-01", "2025-12-31")
df["date"] = pd.to_datetime(
    np.random.choice(date_range, size=len(df))
```

5. Reddit keywords data cleanup





A. The Read and Inspect function groups

pd.read_pickle(): Used to load the dataset from the .pkl format. This is the best way to keep intact Python's complex data types that formats like CSV can lose.

df.info(): This function plays a key role in examining the data set structure, including the number of records, data columns, and corresponding data types (Dtype), helping to identify missing (null) values for processing in a later step.

B. Data Cleaning function group

C. **Df.dropna()**: Removes rows that contain empty (NaN) values. This ensures data integrity, avoiding errors for future analytical or machine learning models. **df.drop_duplicates()**: Eliminates duplicate records, ensuring each Reddit comment or post is unique in the dataset.

D. Logical Processing and Data Enrichment Function Group

np.random.choice(): Used twice for data simulation purposes:

- Create a simulated rating column from 1 to 5 with a predefined probability (p).

- Generate a random date column based on a set date_range time period.

df.apply(classify_sentiment): This is an important method in Pandas that applies a self-defined function (classify_sentiment) to each line of data in a ratingcolumn.

Functional logic: Classify emotions into **Positive** (>=4), **Neutral** (==3), and

Negative (<3).

E. Format and Publish function groups

pd.date_range(): Establishes a timeframe from the beginning of 2024 to the end of 2025, serving the creation of timeseries data. **df.rename()**: Normalizes the names of the attributes (text to text_content, date to review_date). This step is important to ensure that the column name matches the schema of the target database exactly.

df.to_csv(): Converts the DataFrame object into a physical file .csv. The index=False parameter is used to remove Pandas' default index column, making the file cleaner.

6. SQL code to create databases and tables

```sql
CREATE DATABASE reddit_bi;

USE reddit_bi;

CREATE TABLE reddit_reviews (
    review_id INT AUTO_INCREMENT PRIMARY KEY,
    text_content TEXT,
    subreddit VARCHAR(255),
    sentiment VARCHAR(50),
    rating INT,
    review_date DATETIME
);
```

After creating the table, the data from the reddit_cleaned.csv file is imported into the reddit_reviews table using the table data import wizard tool in MySQl workbench.

7. Code SQL test and analyze data

7.1. Check the number of records

```sql
SELECT COUNT(*) AS total_reviews
FROM reddit_reviews;
```

7.2. View sample data

```sql
SELECT *
FROM reddit_reviews
LIMIT 10;
```

7.3. Review statistics by sentiment

```sql
SELECT
    sentiment,
    COUNT(*) AS total_reviews
FROM reddit_reviews
GROUP BY sentiment
ORDER BY total_reviews DESC;
```

7.4. Review statistics by rating

```sql
SELECT
    rating,
    COUNT(*) AS total_reviews
FROM reddit_reviews
GROUP BY rating
ORDER BY rating;
```

7.5. Review statistics by year

```sql
SELECT
    YEAR(review_date) AS year,
    COUNT(*) AS total_reviews
FROM reddit_reviews
GROUP BY YEAR(review_date)
ORDER BY year;
```

7.6. Review statistics by subreddit

```sql
SELECT
    subreddit,
    COUNT(*) AS total_reviews
FROM reddit_reviews
GROUP BY subreddit
ORDER BY total_reviews DESC;
```

7.7. Review statistics by keywords

```sql
SELECT keyword, count
FROM reddit_keywords
WHERE subreddit = 'All'
ORDER BY count DESC
LIMIT 10;
```

IV, BUILD A POWER BI DASHBOARD

(We created 2 dashboards, which are reddit social network keywords and reddit social network emotions corresponding to the two folders reddit_keywords and reddit_Emotion in github)

Dashboards are built using Power BI based on processed data. The main components include:

*Reddit_emotion

- **Total Reviews**: The total number of reviews.
- **Total Ratings**: Total Ratings.
- **Sentiment Distribution**: Distribution of user emotions.
- **Reviews by Rating**: The number of reviews by rating.
- **Review Trend Over Time**: The trend in the number of reviews over time.
- **Reviews by Rating & Sentiment**: The relationship between rating and sentiment.
- **Slicer sentiment**: Filter by sentiment.
- **Slicer review_date**: Filter by time.
- *****Reddit_keywords:
- **Bar chat:** the number of keywords searched.
- **Word cloud:** the vocabulary that is searched.
- **Slicer** : filter by search agent.

The dashboard makes it easy for viewers to observe data overviews and analyze user trends on Reddit.

V, INSIGHT ANALYTICS

From the dashboard, some key insights can be drawn:

First, the majority of comments are in **the Positive** category, suggesting that users tend to respond more positively than negatively and neutrally.

Secondly, high ratings, especially 5 ratings, account for a large proportion. This indicates a relatively high level of user satisfaction.

Third, the number of reviews has changed over time. The dashboard shows that review volume tends to decrease slightly between 2024 and 2025.

Fourth, the Reviews by Rating & Sentiment chart shows that high ratings are often associated with positive sentiment, while low ratings are often associated with negative sentiment.

This shows a reasonable relationship between ratings and user emotions.

Fourth, most of the keywords searched are those related to politics and economics, in addition to a few keywords related to life and comedy.

VI, CONCLUSION

The exercise has built a complete data analysis process including Python, MySQL, and Power BI.

Results:

- Processing Reddit data using Python.
- Store data in a MySQL database.
- Query data using SQL.
- Build visual dashboards with Power BI.
- Draw insights from social media data.

Through this article, it can be seen that social network data is a valuable source of information in analyzing user emotions and behavior. The combination of Python, MySQL, and Power BI helps build a relatively complete and scalable data analysis system.

-End